

Supply, Demand, and Market Equilibrium

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A companion to the lecture slides. Each slide that hides a step is reproduced here and worked through line by line.

Before we begin

Required readings

Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 4** — The Market Forces of Supply and Demand.

Read the chapter alongside these slides.

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Slide 2

Working through it

Why this chapter. Mankiw's chapter 4 runs the argument in the order these slides do: define the market, build demand, build supply, put them together, and then work out what happens when something shifts. Everything in this lecture appears there, and the vocabulary matches.

One difference worth knowing before you read. Mankiw works mostly from tables and diagrams. These slides do the same work algebraically first — write the equation, tabulate

it, rearrange it, plot it — and Part 9 closes the loop by going the other way, from a table back to the equation. Neither route is more correct; having both means you can start from whichever the question hands you.

How to use it. Read the chapter once before the material and once after. The slides move faster and leave out the worked detail; the text is slower and does not tell you which steps carry the weight.

Outline

Outline

1. **Markets and competition** — these are the fundamentals.
2. **Demand** — we explain the law of demand, the equation, the table, the curve.
3. **From one buyer to the market** — where a market demand curve comes from.
4. **What shifts demand** — and why a shift is not a movement.
5. **Supply** — the same four steps, from the sellers' side.
6. **What shifts supply** — and the one shifter that runs backwards.
7. **Market equilibrium** — this is where the forces of supply and demand are in balance.
8. **Comparative statics** — explains what happens to equilibrium when things change.
9. **From table to equation to graph** — when you know the data first, not the equation.

Each part ends with a question for you to answer before we move on.

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Slide 3

Working through it

How the nine blocks fit together. The order is not arbitrary, and knowing the shape of it makes the whole thing easier to hold.

Blocks 1–4 build the **demand** side: what a market is, what demand means, where a market demand curve comes from, and what makes it move.

Blocks 5–6 do the same job for **supply**. They go faster, because the method is identical — deliberately so.

Block 7 is where the two halves **meet** and are solved together. Block 8 is what the model is *for*. Block 9 runs the entire method **backwards**, which is what you would actually do with real data.

The recurring pattern. Four steps appear three separate times: write the equation, plug the numbers in, rearrange to plot, draw the graph. Learn them once in Block 2 and Blocks 5 and 9 cost you almost nothing.

1. Part 1: Markets and Competition

The puzzle we are here to solve

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How is the price of a cup of coffee actually determined?

Think about everything that happened before you paid it. Beans planted on a hillside a continent away. A growing season, and weather nobody controls. Pickers hired and paid. Sacks trucked to a port, freight across an ocean, roasting and blending and packing. Then a shop, a machine, a barista. And at the end of all of it, a price.

Thousands of people, none of whom know you, and no one setting the price.

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Slide 5

Working through it

The coffee in front of you took two or three years and several continents to produce. Thousands of people were involved. No two of them faced the same information or the same incentives, and none of them held more than a fragment of the whole. The grower committed money seasons before you drank anything, against a price that did not yet exist.

Now suppose you wanted to plan that process centrally. The planner would need to know the technical conditions at every stage, what every buyer wants, and the opportunity cost of every input — all at once. That knowledge is scattered across thousands of people, and much of it is never written down. The Nobel laureate Friedrich Hayek made this point in 1945: the economic problem is not allocating resources when you already know the facts. It is using knowledge that no single mind holds.

And yet the result looks planned. Prices are roughly the same across a city. Roughly the

right amount is produced. Shops rarely run out. So the question is not why coffee costs four dollars rather than five. It is why there is a settled price at all.

What is a market?

What is a market?

Market. A market is a place or context where buyers and sellers trade. Buyers as a group determine demand; sellers as a group determine supply.

A market need not be a place. Crude oil trades through screens linking continents; consumers buy goods via internet platforms.

What makes it a market is the trading, not the geography.

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Slide 6

Working through it

Reading the definition. A market is a group of buyers and sellers of a particular good. That is deliberately broad, and two things are worth noticing about what it leaves out.

It says nothing about place. A market is not a building. Crude oil trades across continents through screens; haircuts trade within a few streets. Neither is more or less a market than the other.

It says nothing about size. Two buyers and two sellers form a market. So do two million. Size will matter shortly — it is what the next definition turns on — but it is not part of being a market.

The phrase to hold on to. Buyers *as a group* determine demand; sellers *as a group* determine supply. Individual behaviour is the raw material, but the model works with the totals. Block 3 shows exactly how the individuals are added up.

Markets and competition

Markets and competition

Market. A place or context where buyers and sellers meet to trade a particular good or service.

Buyers as a group determine market demand. Sellers as a group determine market supply.

Competition. Rivalry between agents over scarce resources.

Agents here can be individuals, firms, or governments. Buyers compete with each other for coffee. Sellers compete with each other for customers. Out of that rivalry a price emerges — and no individual buyer or seller plans it.

The market price is determined by the interaction of buyers and sellers.

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Slide 7

Working through it

Why this definition carries the whole model. Everything from here on assumes it, so it is worth being precise about what it does and does not say.

What it says. There are so many buyers and sellers that no single one of them can influence the price. Each therefore treats the price as a fact of the world, like the weather — which is what *price taker* means.

Why the shop cannot escape it. Double your price and buyers walk to the shop next door, because the coffee there is much the same. Halve it and you are swamped by more business than you can serve, and you are losing money on every cup. Neither of you is choosing the price; you are discovering it.

What this buys us. It lets us speak of *the* price of coffee, in the singular. Without the assumption there is no such thing — there is a different price at every shop, and the whole diagram falls apart.

What it costs us. It is an assumption, not a fact, and there are real markets where it plainly fails. When a market has one seller, that seller *chooses* the price, and everything in this lecture has to be rebuilt.

2. Part 2: Demand

Quantity demanded

Quantity demanded

Quantity demanded (Q^d). The amount of a good that buyers are willing and able to buy at a given price, in a given period.

Two words carry weight.

Willing rules out what people would like but will not pay for.

Able rules out what they would pay for but cannot afford.

And Q^d is a **flow**: cups per week, not cups.

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Slide 11

Working through it

Two words, doing real work.

Willing rules out what people would enjoy but will not actually pay for. Wanting something is not demand. Demand requires a decision to part with money.

Able rules out what people would happily pay for but cannot afford. A desire without the means behind it does not appear in this variable at all.

Both conditions must hold. Fail either one and the person contributes nothing to quantity demanded, however strongly they feel.

Why “in a given period” matters. Q^d is a **flow**, not a stock. “Thirty thousand cups” is not an answer to anything; “thirty thousand cups *per week*” is. Change the period and every number in this lecture changes with it, which is why the units on our tables read *000s of cups per week* and not merely *000s of cups*.

And note the superscript. Q^d means quantity *demand*ed, which will shortly need distinguishing from Q^s , quantity supplied. They are equal only at one particular price, and finding that price is the business of Block 7.

The law of demand

The law of demand

Law of demand. Other things equal, when the price of a good rises, the quantity demanded of that good falls.

Hold on to the phrase **other things equal**. It is not decoration.

For now: freeze income, tastes, and every other price. Vary only the price of coffee.

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Slide 12

Working through it

Why this is called a law. Not because it cannot be violated, but because it holds so widely that the exceptions are worth naming individually. It is an empirical regularity, observed across goods, countries and centuries.

The phrase that is not decoration. *Other things equal* — in Latin, *ceteris paribus* — is the load-bearing part of the sentence, and it is the reason Block 4 exists.

Here is why it matters. Suppose you observe that the price of coffee rose last year *and* people bought more of it. Does that refute the law of demand? No, because other things were not equal: incomes may have risen at the same time. The law makes a claim about what happens when the price moves **and nothing else does**.

What we are freezing, concretely. Income, tastes, the price of tea, the price of sugar, the number of buyers, and expectations about future prices. All held still. Only the price of coffee is allowed to vary.

Why that is not a cheat. It is the standard scientific move: change one thing at a time so you can attribute the result to it. Block 4 unfreezes those variables one by one and works out what each does.

Step 1 — write the equation**Step 1 — write the equation**

For our coffee market:

$$Q^d = 70 - 10P$$

Q^d in thousands of cups per week. P in dollars per cup.

The 70 — what buyers would take if coffee were free.

The -10 — each extra dollar cuts quantity demanded by 10 thousand cups. *The minus sign is the law of demand, written in algebra.*

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Slide 13

Working through it

Why start with an equation rather than a picture. Because a picture can be drawn sloppily and still look convincing, whereas an equation can be checked. Every claim that follows has to survive being tested against this line.

Reading it aloud. Quantity demanded equals seventy, minus ten times the price.

Where the 70 comes from. It is the quantity buyers would take if coffee were free — set $P = 0$ and $Q^d = 70$. Nobody will ever observe that transaction, which is worth noticing: the equation here is assumed, so that intercept is a feature of the model rather than something anyone observed.

Where the 10 comes from. It is the slope. Every extra dollar of price costs ten thousand cups. Check it against the equation: raise P by one and Q^d falls by exactly ten, whatever price you started from. That constancy is what makes this a straight line.

Why the minus sign is not decoration. It is the law of demand. A positive coefficient there would say that buyers want more as the price rises. The sign is the economics; the ten is only the magnitude.

The units. Q^d in thousands of cups per week, P in dollars per cup. State them or the numbers mean nothing.

Step 2 — plug the numbers in

Step 2 — plug the numbers in			
Price P (\$/cup)	Substitute into $Q^d = 70 - 10P$		Q^d (000s/week)
1	$70 - 10(1)$	=	60
2	$70 - 10(2)$	=	50
3	$70 - 10(3)$	=	40
4	$70 - 10(4)$	=	30
5	$70 - 10(5)$	=	20
6	$70 - 10(6)$	=	10
7	$70 - 10(7)$	=	0

The table is not a separate object. **It is the equation, evaluated seven times.**

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Slide 14

Working through it

What this step actually does. An equation is a machine for generating points: feed it a price, receive a quantity. Doing that seven times produces the schedule.

One row, built from scratch. Take $P = 4$. Substitute into the equation:

$$Q^d = 70 - 10(4) = 70 - 40 = 30$$

Thirty thousand cups per week. Every other row is that same arithmetic with a different price, which is why the middle column shows the substitution rather than hiding it.

The claim underneath the table. The table is not a separate object that happens to agree with the equation. **It is** the equation, evaluated seven times. There is no independent information in it.

Which is why you can check any row. Put a finger on any number in the right-hand column and you can rebuild it from the equation in one line. Nothing here has to be taken on trust, and that is the point of showing the substitutions.

Why seven rows and not seventy. Seven is enough to see the pattern and few enough to read. The equation holds at every price in between, including prices that are not whole dollars.

One row worth pausing on

One row worth pausing on

At $P = 7$, quantity demanded is **zero**. The price is so high that nobody buys at all.

Call this the choke price.

It is where the demand curve meets the price axis, and you can read it straight off the equation: set $Q^d = 0$ and solve for P .

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Slide 15

Working through it

What the last row says. At $P = 7$, quantity demanded is zero. The price has risen so far that nobody buys any coffee at all.

How to find it without a table. Set $Q^d = 0$ and solve for P :

$$70 - 10P = 0 \implies 10P = 70 \implies P = 7$$

Two lines, and no table required. That is the general method, and it works for any linear demand curve you are handed.

Where it sits on the diagram. It is the price at which the demand curve meets the vertical axis — the top end of the curve.

Why it has a name. **Choke price** is the standard term: the price that chokes off demand entirely. It appears again in Block 9, where it turns out to be something the equations can tell you and the raw data cannot.

Remember the manoeuvre, not just the answer. Setting a quantity to zero and solving for price is used twice more in this lecture: once for the shut-down price in Block 5, and again in Block 9.

Step 3 — rearrange to plot**Step 3 — rearrange to plot**

Economists put price on the *vertical* axis. That is backwards from the mathematics you were taught — a habit inherited from Alfred Marshall.

So rearrange to make P the subject:

$$Q^d = 70 - 10P \implies 10P = 70 - Q^d \implies P = 7 - \frac{1}{10}Q^d$$

Same relationship. Same information. Different subject.

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Slide 16

Working through it

Why this step is necessary at all. Purely convention, and an inconvenient one.

In mathematics, the variable you choose goes on the horizontal axis and the variable that responds goes on the vertical. Buyers choose in response to price, so price ought to be horizontal. Economists put it vertical instead. We inherited that from Alfred Marshall in 1890 and have been stuck with it ever since.

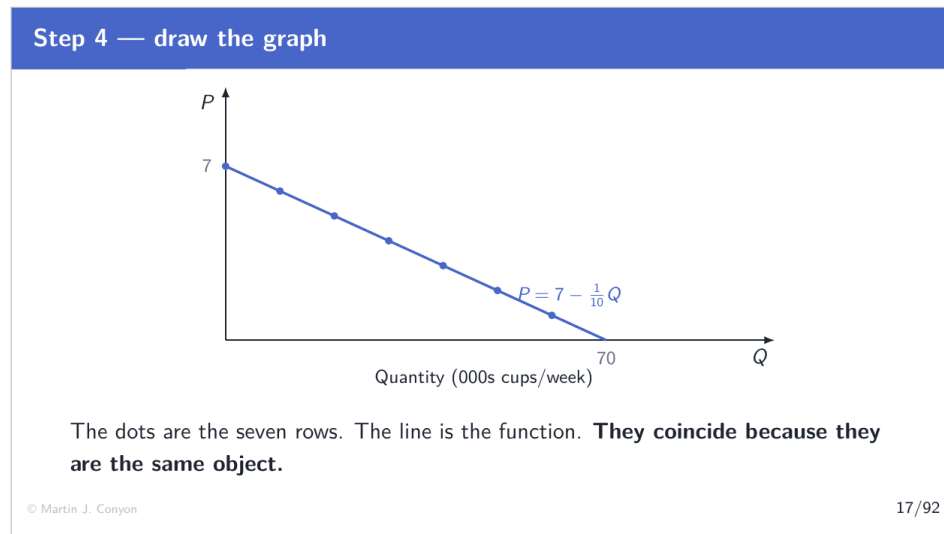
So the equation has to be turned around. Follow the algebra across the slide:

$$Q^d = 70 - 10P \implies 10P = 70 - Q^d \implies P = 7 - \frac{1}{10}Q^d$$

Step one moves $10P$ to the left and Q^d to the right. Step two divides everything by ten.

Check the two numbers against what you already know. The 7 is the choke price — set $Q^d = 0$ and $P = 7$. The $\frac{1}{10}$ is the reciprocal of the original slope of 10, which is what you would expect from turning the relationship inside out.

Nothing economic has happened. Same relationship, same information, different subject. This form is sometimes called *inverse demand*.

Step 4 — draw the graph

Slide 17

Working through it

What is on the axes. Quantity horizontally, price vertically — the convention from the previous step.

What the dots are. The seven rows of the table, plotted. The row $P = 4$, $Q = 30$ becomes the point 30 across and 4 up.

What the line is. The function $P = 7 - \frac{1}{10}Q$, drawn continuously.

Why they coincide. Because they are the same object. The dots are the function evaluated at seven points; the line is the function evaluated everywhere. If a dot sat off the line, one of them would be wrong.

Reading the two labelled numbers. The 70 on the horizontal axis is the quantity at a price of zero. The 7 on the vertical axis is the choke price. Those are the two ends of the curve, and both came out of the equation rather than off a ruler.

The claim this frame is really making. A demand curve is not a downward-sloping shape drawn by feel. It is the graph of one specific function, and every later claim about it must survive being checked against that function. That discipline is what separates economics from storytelling with pictures.

3. Part 3: From One Buyer to the Market

Where did that curve come from?

Where did that curve come from?

We wrote down a *market* demand curve and never asked where it came from.

A market demand curve is built from individual demand curves.

Each buyer has their own. To get the market curve, add up the quantities **at each price**.

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Slide 21

Working through it

The hole this block fills. We wrote down $Q^d = 70 - 10P$ and treated it as given. But a market demand curve is not a primitive object — it is assembled, from the demand curves of the individual people in the market.

The rule, and the trap inside it. To build the market curve, add up the quantities **at each price**.

Read that twice. We add *quantities*, holding the price fixed. We never add prices. Adding prices would be asking what two buyers would jointly pay, which is a different question and not one this diagram answers.

Why the phrase “at each price” does the work. Every buyer faces the same market price — that is what the competitive assumption from Block 1 bought us. So fix a price, ask each buyer how much they want at it, and total the answers. Then repeat for the next price.

Two buyers

Two buyers

Suppose the market has just two buyers, Ama and Ben:

$$Q_{\text{Ama}}^d = 42 - 6P, \quad Q_{\text{Ben}}^d = 28 - 4P$$

Ama buys more at every price below \$7 and reacts more strongly to price changes (slope -6 against -4).

Both stop buying at \$7 — they happen to share a choke price.

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Slide 22

Working through it

Why only two buyers. To make the arithmetic visible. A real market has thousands, and the principle is identical — there are simply more terms in the sum.

Reading the two equations. Ama is $42 - 6P$; Ben is $28 - 4P$. Both slope downward, as the law of demand requires, but they differ in two ways.

Difference one: how much. Ama's intercept is larger, 42 against 28, so she buys more at every price below \$7, where both still buy. At $P = 3$: Ama takes $42 - 18 = 24$ while Ben takes $28 - 12 = 16$.

Difference two: how responsively. Ama's slope is 6 against Ben's 4, so each extra dollar costs Ama six units and Ben only four. Ama reacts more strongly.

The convenient coincidence. Both choke at \$7. Check it: Ama, $42/6 = 7$; Ben, $28/4 = 7$. Nothing forces that, and it is arranged here to keep the picture clean. Two buyers with different choke prices produce a market curve with a **kink** at the higher of the two, where one buyer drops out and the other carries on alone.

Add the quantities at each price

Add the quantities at each price				
P	$Ama = 42 - 6P$	$Ben = 28 - 4P$	Sum	Market Q^d
1	36	24	$36 + 24 = 60$	60
2	30	20	$30 + 20 = 50$	50
3	24	16	$24 + 16 = 40$	40
4	18	12	$18 + 12 = 30$	30
5	12	8	$12 + 8 = 20$	20
6	6	4	$6 + 4 = 10$	10
7	0	0	$0 + 0 = 0$	0

The last column is the market demand curve we started with.

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Slide 23

Working through it

Building one row from scratch. Take $P = 3$.

Ama: $42 - 6(3) = 42 - 18 = 24$. Ben: $28 - 4(3) = 28 - 12 = 16$.

Sum: $24 + 16 = 40$.

That is the whole operation, and every row is that same arithmetic at a different price.

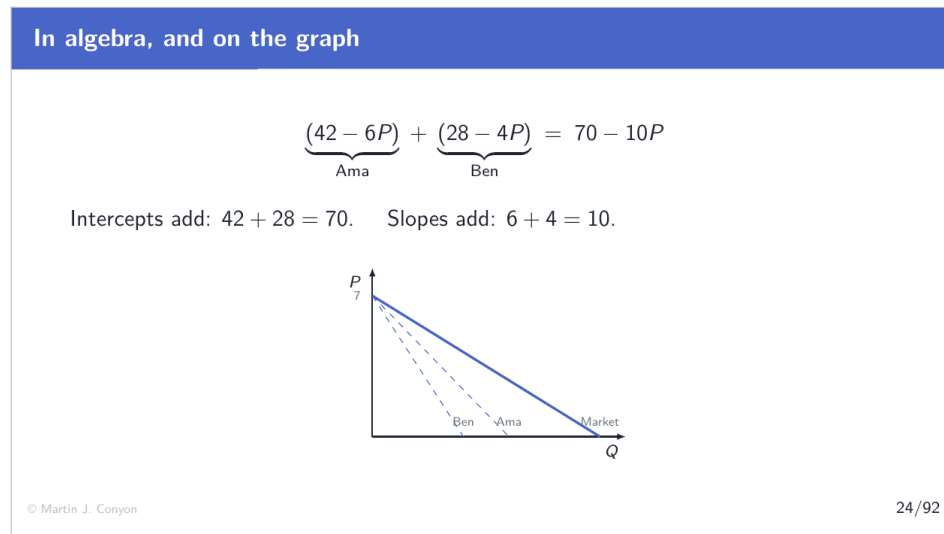
Now the point of the table. Read down the final column: 60, 50, 40, 30, 20, 10, 0.

Now go back to the demand schedule built in Block 2 from $Q^d = 70 - 10P$. It reads 60, 50, 40, 30, 20, 10, 0.

Identical. The market curve we were handed at the start of this lecture was these two buyers all along.

Why that matters. It means the market demand curve is not an assumption about markets. It is a consequence of assumptions about people. Everything that happens to it later — every shift in Block 4 — is ultimately something happening to individuals.

In algebra, and on the graph



Slide 24

Working through it

The addition, done once instead of seven times.

$$(42 - 6P) + (28 - 4P) = (42 + 28) - (6 + 4)P = 70 - 10P$$

Collect the constants and collect the P terms. That is all.

Which gives the rule. For linear demand curves, **intercepts add and slopes add**. $42 + 28 = 70$; $6 + 4 = 10$. One line of algebra replaces the whole table.

What the figure shows. Quantity horizontally, price vertically. The two dashed lines are Ama and Ben; the solid line is the market.

How to read it. Pick any height — any price — and look across. The market curve sits to the right of both individual curves, by exactly the sum of their two quantities. That sideways addition is why this is called **horizontal summation**.

Why all three meet at 7. Because both buyers happen to choke at \$7. Above that price nobody demands anything, so all three curves have run out together.

What would happen otherwise. If Ben choked at \$5 and Ama at \$7, then between \$5 and \$7 the market curve would be Ama's curve, and at \$5 it would kink outward as Ben rejoined.

The market curve would still be a sum — just no longer a straight line.

4. Part 4: What Shifts Demand

We froze the world. Now unfreeze it.

We froze the world. Now unfreeze it.

The curve was drawn holding everything except the price of coffee constant.

Release one of those things and **the curve itself moves**.

Which things, in which direction, and why?

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Slide 28

Working through it

What is about to be undone. Every curve so far was drawn under *other things equal* — income, tastes and all other prices held still. That was necessary to isolate the effect of price. It is also unrealistic, and this block repairs it.

The distinction the block exists to establish. When the price of coffee changes, you **move along** the curve. When anything else changes, the curve itself **shifts**.

Those are different events with different consequences, and confusing them is the most common single error in introductory economics. By the end of this block you will be able to tell them apart by looking at an equation rather than by remembering a rule.

The three questions. Which things move the curve; in which direction; and — the one that actually matters — why.

The intercept is not a fact about coffee

The intercept is not a fact about coffee

Write demand out with the other influences visible:

$$Q^d = 50 - 10P + 6P_T - 4P_S + 2Y$$

P_T = price of tea P_S = price of sugar Y = income index

At today's values — tea \$3, sugar \$2, income 5:

$$50 + 6(3) - 4(2) + 2(5) = 50 + 18 - 8 + 10 = 70$$

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Slide 29

Working through it

The claim. That 70 looks like a fact about coffee. It is not. It is a sum, and the things being summed are exactly the variables we froze.

The fuller equation, read aloud. Quantity demanded equals fifty, minus ten times the price of coffee, plus six times the price of tea, minus four times the price of sugar, plus two times the income index.

Where each coefficient's sign comes from. -10 on coffee's own price is the law of demand. $+6$ on tea is positive because tea is a substitute. -4 on sugar is negative because sugar is a complement. $+2$ on income is positive because coffee is a normal good here. Each sign is an economic claim, not a technical detail — and the next few frames prove each one.

Assembling the intercept. At today's values — tea \$3, sugar \$2, income index 5:

$$50 + 6(3) - 4(2) + 2(5) = 50 + 18 - 8 + 10 = 70$$

So 70 was never a primitive. It is $50 + 18 - 8 + 10$. Change any term in that sum and the intercept changes, which moves the entire curve. That is what a shift *is*, seen from inside the algebra.

The rule in one line

The rule in one line

Changing P moves you *along* the curve — P sits in the equation multiplied by its own slope.

Changing P_T , P_S **or** Y changes the *intercept*, which relocates the whole curve.

That is the entire distinction, and you can see it in the algebra.

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Slide 30

Working through it

Look at where each variable sits in the equation, because the position is the answer.

$$Q^d = \underbrace{50 + 6P_T - 4P_S + 2Y}_{\text{the intercept}} - 10 \underbrace{P}_{\text{own price}}$$

Own price. P is multiplied by its own slope. Change it and you compute a different point *on the same line*. That is a **movement along** the curve.

Everything else. P_T , P_S and Y all sit inside the intercept. Change any of them and the intercept changes, which relocates the whole line. That is a **shift**.

Why this is better than a rule to memorise. A memorised rule fails on an event you have not seen. This does not: given any event, ask which term of the equation it touches. If it touches P , you move along. If it touches the intercept, the curve shifts.

The examinable sentence. A change in the good's own price moves you along the curve. A change in anything else shifts it.

Substitutes

Substitutes

Substitutes. Two goods that serve the same purpose, so that a rise in the price of one increases demand for the other. The coefficient on the other good's price is *positive*.

Worked example. A blight hits the tea crop; tea goes \$3 → \$5.

$$50 + 6(5) - 4(2) + 2(5) = 82 \quad \Rightarrow \quad Q^d = 82 - 10P$$

Demand rose by 12 thousand cups *at every price* — and the price of coffee never changed.

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Slide 31

Working through it

The definition is a sign, not a story. Two goods are substitutes when the coefficient on the other good's price is **positive**. In our equation the coefficient on P_T is +6, and that is what makes tea a substitute for coffee.

Why positive means substitute. A positive coefficient says: tea's price up, coffee's quantity up. Buyers move *toward* coffee when tea gets dearer, which is what serving the same purpose means.

The worked example, line by line. Blight hits the tea crop and tea goes from \$3 to \$5. Only P_T changes; everything else holds.

$$50 + 6(5) - 4(2) + 2(5) = 50 + 30 - 8 + 10 = 82$$

So the new demand equation is $Q^d = 82 - 10P$.

How big is the shift? The intercept rose from 70 to 82, so demand is up by 12 thousand cups **at every price**. Check it: at \$4, demand was $70 - 40 = 30$ and is now $82 - 40 = 42$. Up 12. At \$6: was 10, now 22. Up 12 again.

The critical observation. The price of coffee never changed. Not once. We shifted the curve; we did not slide along it. If you can say why that is true here, you have the distinction.

Complements

Complements

Complements. Two goods consumed together, so that a rise in the price of one decreases demand for the other. The coefficient is *negative*.

Worked example. Sugar goes \$2 → \$4.

$$50 + 6(3) - 4(4) + 2(5) = 62 \quad \Rightarrow \quad Q^d = 62 - 10P$$

Demand fell by 8 thousand cups at every price.

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Slide 32

Working through it

Again the definition is a sign. Two goods are complements when the coefficient on the other good's price is **negative**. Ours on P_S is -4 .

Why negative means complement. A negative coefficient says: sugar's price up, coffee's quantity down. The two move in *opposite* directions, which is what being consumed together implies.

The arithmetic. Sugar goes from \$2 to \$4:

$$50 + 6(3) - 4(4) + 2(5) = 50 + 18 - 16 + 10 = 62$$

So $Q^d = 62 - 10P$, and the intercept fell from 70 to 62 — demand down by 8 thousand cups at every price.

Check the size against the coefficient. Sugar rose by \$2 and the coefficient is -4 , so the intercept should fall by $4 \times 2 = 8$. It did. That check works for every shift in this block, and it is worth doing.

Why the next frame exists. The arithmetic here is easy. The *reasoning* is the thing students reverse more often than anything else in this lecture, so it gets a frame to itself.

The complement chain, stated slowly

The complement chain, stated slowly

Sugar and coffee are consumed **together**. What a buyer really chooses is *the pair*.

When sugar becomes dearer, **the pair** becomes dearer — even though coffee itself has not changed price at all.

Buyers buy fewer coffee-and-sugar combinations, which means less coffee.

Demand for coffee shifts left.

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Slide 33

Working through it

The chain, in order, with nothing skipped.

One. Sugar and coffee are consumed together. What the buyer is really choosing is not coffee alone — it is the *pair*.

Two. Sugar becomes dearer. Therefore the pair becomes dearer.

Three. And note what has *not* happened: coffee's own price has not changed by a single cent.

Four. Facing a dearer pair, buyers buy fewer pairs.

Five. Fewer pairs means less coffee, at every coffee price.

Six. So the coffee demand curve shifts *left*.

Which step do people drop? Step three. Having been told that something got more expensive, the instinct is to slide leftward *along* the coffee demand curve — which would be right if coffee's price had risen. It did not. Sugar's did. So the curve moves instead.

The test that catches it. Ask: did the price of **coffee** change? Here, no. So this cannot be a movement along the coffee curve, whatever else it is.

Income: normal and inferior goods

Income: normal and inferior goods

Normal good. Demand increases when income rises. The coefficient on Y is positive.

Inferior good. Demand decreases when income rises. The coefficient on Y is negative.

Ours is $+2$, so coffee is normal here. Income $5 \rightarrow 10$ gives an intercept of 80.

Inferior does not mean bad. It means people buy less of it as they get richer: instant coffee, coach travel, own-brand cereal.

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Working through it

Two categories, distinguished by a sign. Positive coefficient on income: **normal** good, bought more of as income rises. Negative coefficient: **inferior** good, bought less of.

Ours is $+2$, so coffee is a normal good in this market.

The worked shift. Income index from 5 to 10:

$$50 + 6(3) - 4(2) + 2(10) = 50 + 18 - 8 + 20 = 80$$

The intercept rises from 70 to 80. Check against the coefficient: income rose by 5, the coefficient is $+2$, so the rise should be $2 \times 5 = 10$. It is.

The word “inferior” is badly chosen. It does not mean poor quality. It means people move *away* from the good as they get richer, having found something they prefer: instant coffee, coach travel, supermarket own brand. Many inferior goods are perfectly good.

And the category is not fixed to the good. Whether something is inferior is a question about a particular population at a particular time, not a property stamped on the item. Bus travel is inferior in wealthy cities and normal in poorer ones — same service, opposite classification.

The full list of demand shifters

The full list of demand shifters		
Change	Effect on demand	Curve
Income rises (normal good)	increases	shifts right
Income rises (inferior good)	decreases	shifts left
Price of a substitute rises	increases	shifts right
Price of a complement rises	decreases	shifts left
Tastes turn toward the good	increases	shifts right
More buyers enter the market	increases	shifts right
Buyers expect a higher future price	increases	shifts right

The last row is the only forward-looking one: expectations shift demand *now*, on a price change that has not happened yet.

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Working through it

How to use this table. Learn the logic, not the rows. Every entry works the same way: it changes a term inside the assembled intercept, which moves the whole curve. Understand that and you can handle shifters that are not listed.

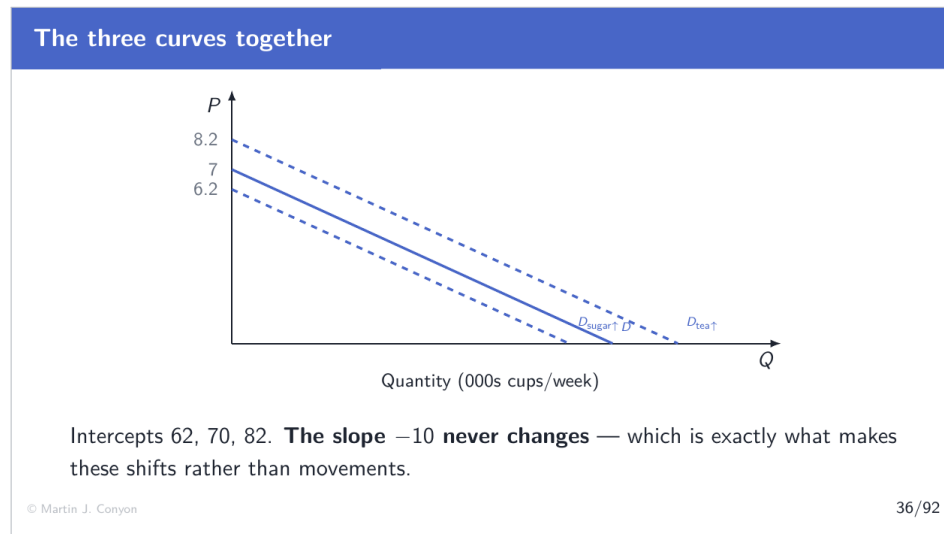
Checking two rows against the equation. “Price of a substitute rises — shifts right”: the coefficient is positive, so a higher P_T raises the intercept. Confirmed by the tea example, $70 \rightarrow 82$. “Price of a complement rises — shifts left”: negative coefficient, lower intercept, $70 \rightarrow 62$. Confirmed.

The two rows not in our equation. **Tastes** and **number of buyers** are not written into $50 - 10P + 6P_T - 4P_S + 2Y$, but they belong on the list. More buyers means more total demand at every price — which is Block 3’s horizontal summation with extra people added to the sum.

The last row is different in kind. Every other entry is about conditions *now*. Expectations are about conditions *later*: buyers expecting a higher price next month buy more today.

So the curve moves now, on the strength of a price change that has not happened and may never happen. That is worth sitting with, because it means today’s price depends partly on what people believe about tomorrow.

The three curves together



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Working through it

What is on the axes. Quantity horizontally, price vertically, as always. Three demand curves rather than one.

Which is which. The solid middle line is the original, intercept 70. The dashed line on the left is after sugar rose, intercept 62. The dashed line on the right is after tea rose, intercept 82. All three came from the arithmetic on the previous frames.

Reading the vertical axis. The three labels 6.2, 7 and 8.2 are the choke prices — where each curve meets the price axis. Check one: for $Q^d = 82 - 10P$, set $Q^d = 0$ and $P = 8.2$. The shifts move the choke price too.

The thing to notice, and it is the point of the frame. The three lines are **parallel**.

Why they must be. The slope is -10 in all three equations. None of these events changed *how strongly* buyers react to the price of coffee — only how much they want at each price. A shift moves the line; it does not tilt it.

And that is the definition of a shift, seen geometrically. If a line had tilted, something would have changed buyers' responsiveness to coffee's own price, and that is a different kind of event altogether.

5. Part 5: Supply

Quantity supplied

Quantity supplied

Quantity supplied (Q^s). The amount of a good that sellers are willing and able to sell at a given price, in a given period.

Law of supply. Other things equal, when the price of a good rises, the quantity supplied of that good rises.

Note it runs the *opposite* way to demand.

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Working through it

Read the two definitions against their demand counterparts, because the parallel is exact and it is the reason this block moves quickly.

Quantity supplied. The amount sellers are **willing and able** to sell at a given price, in a given period. The same two words carry the same weight: willing rules out sellers who could produce but will not at that price; able rules out those who would but cannot. And it is a **flow** again — cups per week.

The law of supply. Other things equal, price up, quantity supplied *up*.

The one difference, and it is the important one. Demand runs downward; supply runs upward. Buyers and sellers respond to the same price movement in opposite directions — and that opposition is exactly what makes an equilibrium possible. Curves that both slope the same way need not cross at all.

The superscript. Q^s , quantity supplied, as against Q^d , quantity demanded. Keeping them distinct matters from Block 7 onward, where the whole exercise is finding the single price at which they are equal.

Why supply slopes upward

Why supply slopes upward

The reasoning runs through **cost**.

Producing more coffee means longer hours, extra staff, beans from less convenient places.

All of that costs more per cup.

A higher price is what makes those extra costs worth bearing.

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Working through it

Why justify the slope at all? Because a curve you cannot explain is a curve you will draw the wrong way round under pressure. The reasoning runs through **cost**.

The mechanism, step by step. To produce more coffee a shop must stay open longer, pay staff for weekend shifts, and buy beans from suppliers it would not have used first. Each of those raises the cost of producing an *additional* cup.

So the cost of the next cup rises as output rises. That is the crucial sentence. It is not that coffee is expensive; it is that each successive cup costs more than the one before it.

Which is why a higher price brings out more. At \$1 a cup, only the cheapest production is worth doing. At \$5, production that costs \$4 a cup becomes worth doing as well, so more appears.

The parallel with demand. Demand slopes down because buyers value additional cups less and less. Supply slopes up because sellers find additional cups cost more and more. Both curves are really statements about *the next unit*, which is a habit of thought economics returns to constantly.

Step 1 — write the equation

Step 1 — write the equation

$$Q^s = 10P - 10$$

The +10 — each extra dollar brings out another 10 thousand cups. Positive, because that is the law of supply.

But the intercept is -10 . What is a negative quantity?

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Working through it

Same step one as the demand side. Write it as an equation, because an equation can be checked.

Reading it aloud. Quantity supplied equals ten times the price, minus ten.

The slope, +10. Every extra dollar brings out another ten thousand cups. Check it: raise P by one and Q^s rises by exactly ten, from any starting price. And the *plus* sign is the law of supply written in algebra, exactly as the minus sign was the law of demand.

The intercept, -10 , and why it should worry you. Substituting $P = 0$ gives $Q^s = -10$. A negative quantity supplied is meaningless — nobody supplies minus ten thousand cups of anything.

Do not dismiss it as a technicality. An equation that returns nonsense at some values is telling you something about where it applies. The next frame reads that message properly, and the answer turns out to be economically useful rather than merely tidy.

Reading the negative intercept

Reading the negative intercept

It does **not** mean sellers supply minus ten thousand cups at a price of zero. That would be meaningless.

It means the curve reaches the axis at a *positive price*. Set $Q^s = 0$:

$$10P - 10 = 0 \implies P = 1$$

Below \$1 a cup, nothing is supplied at all. Call \$1 the **shut-down price**.

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Working through it

What the negative intercept does not mean. It does not mean sellers supply a negative quantity at a price of zero. That is meaningless.

What it does mean. It means the supply curve reaches the horizontal axis at a **positive price**. Set $Q^s = 0$ and solve:

$$10P - 10 = 0 \implies 10P = 10 \implies P = 1$$

So below \$1 a cup, nothing is supplied at all. Not a little. Nothing. At that price no seller thinks production is worth the cost, which is exactly the argument from the previous frame.

The name. **Shut-down price:** the price below which sellers shut down.

Notice the method, because you have used it before. Set quantity to zero, solve for price. That is precisely how we found the choke price of \$7 on the demand side. Same manoeuvre, opposite end of the market: the choke price is where buyers stop buying, the shut-down price where sellers stop selling.

And the practical consequence. The equation is only economically meaningful for $P \geq 1$. Below that it returns numbers, and they are not answers. Knowing that is your job; the

algebra will not tell you.

Step 2 — plug the numbers in

Step 2 — plug the numbers in			
Price P (\$/cup)	Substitute into $Q^s = 10P - 10$		Q^s (000s/week)
1	$10(1) - 10$	=	0
2	$10(2) - 10$	=	10
3	$10(3) - 10$	=	20
4	$10(4) - 10$	=	30
5	$10(5) - 10$	=	40
6	$10(6) - 10$	=	50
7	$10(7) - 10$	=	60

Step 3 — rearrange to plot: $P = 1 + \frac{1}{10}Q^s$

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Working through it

Steps two and three, and there is deliberately nothing new here.

One row from scratch. Take $P = 4$:

$$Q^s = 10(4) - 10 = 40 - 10 = 30$$

Thirty thousand cups per week.

Notice the first row. At $P = 1$, $Q^s = 0$ — the shut-down price, appearing in the table exactly where the algebra said it would.

And notice the direction of travel. On the demand table, quantity fell as price rose: 60, 50, 40. Here it rises: 0, 10, 20. Same format, opposite slope, and that opposition is what will make Block 7 work.

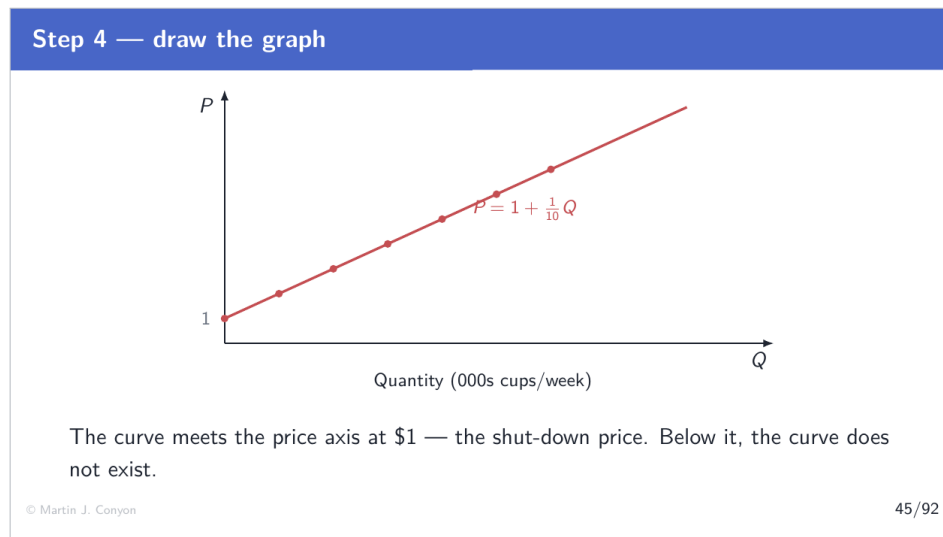
Step three, the rearrangement. Make P the subject so the curve can be plotted with price vertical:

$$Q^s = 10P - 10 \implies 10P = Q^s + 10 \implies P = 1 + \frac{1}{10}Q^s$$

Read the two numbers: the 1 is the shut-down price, and $\frac{1}{10}$ is the reciprocal of the slope, exactly as on the demand side.

The point of this frame. One method, applied twice. If steps one to four felt like four separate things in Block 2, this is where they should start feeling like one.

Step 4 — draw the graph



Slide 45

Working through it

What is on the axes. Quantity horizontally, price vertically, as always.

Dots and line. The dots are the seven rows of the supply table; the line is $P = 1 + \frac{1}{10}Q$. They coincide because they are the same object.

Which way it slopes, and why. Upward, left to right — the law of supply made visible. Compare it with the demand figure from Block 2, which fell.

Where it meets the vertical axis. At \$1, the shut-down price.

And what is deliberately not drawn. There is no curve below \$1. That is not the picture running out of room: below the shut-down price the curve does not exist, because nothing is supplied. A supply curve genuinely begins somewhere.

Compare with demand, which ends somewhere. The demand curve stops at the choke price

of \$7 because above it nobody buys. Both curves occupy a finite range of prices, and the two ranges overlap between \$1 and \$7 — which is where the equilibrium is going to have to be.

6. Part 6: What Shifts Supply

The same logic, mirrored

The same logic, mirrored

The intercept -10 is not a primitive either. It is assembled from the conditions sellers face:

$$Q^s = 10P + 16 - 5P_B + 4T$$

P_B = price of beans (an input) T = index of roasting technology

At beans \$6, technology 1:

$$16 - 5(6) + 4(1) = 16 - 30 + 4 = -10$$

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Working through it

This block is Block 4 again, with sellers. The goal is that you see it as the same idea, not a second one to memorise.

The claim. That intercept of -10 is not a primitive either. It is assembled from the conditions sellers face.

The fuller equation, read aloud. Quantity supplied equals ten times the price, plus sixteen, minus five times the price of beans, plus four times the technology index.

What the new variables are. P_B is the price of beans, an **input** — something sellers must buy in order to produce. T is an index of roasting technology, where a higher number means a better process.

Assembling the intercept. At today's values — beans \$6, technology index 1:

$$16 - 5(6) + 4(1) = 16 - 30 + 4 = -10$$

So -10 was never a primitive either. It is $16 - 30 + 4$. Change the bean price or the technology and that sum changes, which moves the whole supply curve — and, as we shall see, moves the shut-down price with it.

Read the two signs

Read the two signs

Beans, -5: negative, because beans are a *cost*. When an input gets dearer, the margin on each cup falls, the least efficient sellers stop producing, and supply shifts **left**.

Every input behaves this way — wages, rent, energy, shipping.

Technology, +4: positive, because better technology cuts the cost of each cup. Same beans, same hours, more output. Supply shifts **right**.

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Working through it

Each sign is an economic claim. Read them one at a time.

*Beans, -5: negative, because beans are a **cost**.* Follow the chain. Beans get dearer, so the margin on each cup gets thinner, so the sellers who were only just breaking even stop producing, so less is supplied at every price. The curve shifts **left**.

And every input behaves this way. Wages, rent, energy, shipping, insurance. If it is something the seller must buy in order to produce, its coefficient is negative. That generalisation is worth more than the specific number.

Technology, +4: positive, because better technology cuts the cost per cup. Same beans, same

hours, more output — so at any given price it is worth producing more. The curve shifts **right**.

The unifying test. Every supply shifter reduces to one question: does this make it *more* or *less* expensive to produce a cup? More expensive shifts supply left; less expensive shifts it right. That single test handles events that appear on no list.

Two worked examples

Two worked examples

A poor harvest. Beans \$6 → \$8:

$$16 - 5(8) + 4(1) = -20 \quad \Rightarrow \quad Q^s = 10P - 20$$

Supply fell by 10 at every price, *and* the shut-down price rose \$1 → \$2.

Better technology. Index 1 → 3:

$$16 - 5(6) + 4(3) = -2 \quad \Rightarrow \quad Q^s = 10P - 2$$

Supply rose by 8 at every price; the shut-down price fell to \$0.20.

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Working through it

First example, line by line. A poor harvest pushes beans from \$6 to \$8. Only P_B changes:

$$16 - 5(8) + 4(1) = 16 - 40 + 4 = -20$$

So $Q^s = 10P - 20$. The intercept fell from -10 to -20 , which is 10 thousand cups less at every price. Check the size against the coefficient: beans rose \$2, the coefficient is -5 , so the fall should be $5 \times 2 = 10$. It is.

The second effect, which is easy to miss. The shut-down price moved. Set $Q^s = 0$: $10P - 20 = 0$, so $P = 2$. It was \$1. Sellers now need **two** dollars a cup before any coffee appears at all.

Second example, the reverse. The technology index goes from 1 to 3:

$$16 - 5(6) + 4(3) = 16 - 30 + 12 = -2$$

So $Q^s = 10P - 2$, up 8 at every price — and again check it: technology rose by 2, coefficient +4, so $4 \times 2 = 8$.

And its shut-down price. $10P - 2 = 0$ gives $P = 0.20$. Twenty cents. Better technology does not merely raise supply; it lets production happen at prices that previously made it impossible.

The full list of supply shifters

The full list of supply shifters		
Change	Effect on supply	Curve
Input prices rise (beans, wages, energy)	decreases	shifts left
Technology improves	increases	shifts right
More sellers enter the market	increases	shifts right
Sellers leave the market	decreases	shifts left
A tax per unit is imposed	decreases	shifts left
A subsidy per unit is paid	increases	shifts right
Sellers expect a higher future price	decreases	shifts left
Bad harvest, drought or disruption	decreases	shifts left

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Working through it

How to use it. As on the demand side: learn the logic, not the rows. Every entry reduces to the single test from two frames ago — does this make a cup more or less expensive to produce?

Working through four of them with that test. Input prices rise: more expensive, shifts left. Technology improves: less expensive, shifts right. A tax per unit: the seller must now hand over money on every cup, which is exactly like a higher input cost, so left. A subsidy per unit: the reverse, so right.

The two rows about entry and exit. More sellers means more supplied at every price; sellers leaving means less. These are the supply-side counterpart of the “more buyers” row on the demand list, and both are horizontal summation with more or fewer terms in the sum.

Bad harvest, drought, disruption. All the same thing: production becomes harder, so it becomes more expensive, so supply shifts left.

And one row on that list behaves differently from all the others. It gets its own frame next, because it is the one students reverse.

Expectations run the other way

Expectations run the other way

For *buyers*, expecting a higher price next month means buying **more today** — demand shifts **right**.

For *sellers*, expecting a higher price next month means holding stock back — supply shifts **left**.

Buyers pull purchases forward. Sellers push sales back.

Both require a good that can be stored, or a purchase that can be moved in time. Nobody holds back a brewed coffee.

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Working through it

Why this needs its own frame. It is the one shifter that points in *opposite directions* on the two sides of the market, and it is reversed every year.

Buyers. A buyer who expects a higher price next month buys **more today**, to get in before the rise. Demand shifts **right**.

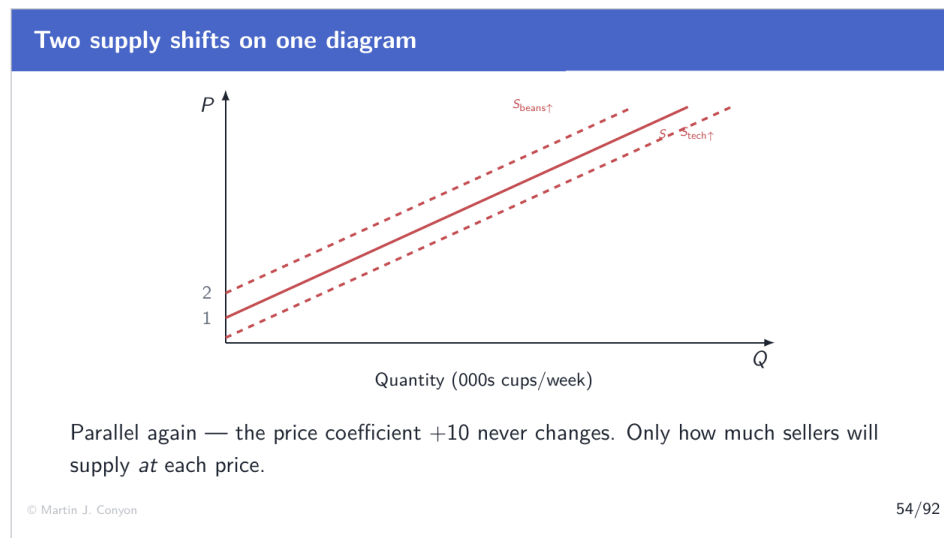
Sellers. A seller who expects a higher price next month holds stock **back**, to sell into the better price later. Supply shifts **left**.

The sentence to memorise. Buyers pull purchases forward; sellers push sales back.

Why they run opposite ways. Both sides are doing the same sensible thing: trading at the better price. But “the better price” means the low one for a buyer and the high one for a seller, so the same expectation moves them in opposite directions.

And notice what this does to the price now. Demand shifting right and supply shifting left both push the current price *up*. So an expectation of a higher price next month tends to raise the price today — the belief starts making itself true before the event it predicts has happened.

Two supply shifts on one diagram



Slide 54

Working through it

What is on the axes. Quantity horizontally, price vertically. Three supply curves, all upward-sloping.

Which is which. The solid middle line is the original, intercept -10 . The dashed line up and to the *left* is after beans got dearer, intercept -20 . The dashed line down and to the *right* is after better technology, intercept -2 .

Reading the vertical axis. The labels 1 and 3 are shut-down prices. The original curve leaves the axis at \$1; the dearer-beans curve at \$2, and the picture is drawn far enough up to show how the whole family climbs.

The thing to notice. **Parallel again.** The slope is +10 in all three equations, exactly as the three demand curves shared a slope of -10 .

Why they must be. None of these events changed *how strongly* sellers respond to the price of coffee. What changed is how much they will supply at each price. The line moves; its steepness does not.

Which is the same test as on the demand side. A shift moves a curve. A change in the good's own price moves you along it. Nothing here tilted anything.

7. Part 7: Equilibrium

Putting the two halves together

Putting the two halves together

Equilibrium. The price and quantity at which quantity demanded equals quantity supplied. The market clears: every buyer willing to pay that price finds a seller, and every seller willing to accept it finds a buyer.

Two curves. One price where they meet.

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Working through it

What has been true so far, and what changes now. We have two curves drawn on the same pair of axes that have so far had nothing to do with each other. Demand was built in Blocks 2–4; supply in Blocks 5–6. Neither has yet said anything about what the price will actually be.

Reading the definition. Equilibrium is a price at which quantity demanded equals quantity supplied: $Q^d = Q^s$.

What “the market clears” means. Every buyer willing to pay that price finds a seller, and every seller willing to accept it finds a buyer. Nobody is left holding coffee they cannot sell, and nobody is left wanting coffee they cannot buy.

Two curves, one crossing point. Demand slopes down, supply slopes up, so they cross exactly once. That is not a drawing convenience — it is why the opposite slopes mattered back in Block 5.

And the harder question this block must also answer. Finding the crossing point is arithmetic. Explaining why a market that starts *somewhere else* should ever arrive there is the real work, and it takes four more frames.

Solving for equilibrium

Solving for equilibrium

Set $Q^d = Q^s$ and solve:

$$\begin{aligned}70 - 10P &= 10P - 10 \\70 + 10 &= 10P + 10P \\80 &= 20P \\P^* &= 4\end{aligned}$$

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Working through it

The whole method in one sentence. Equilibrium means $Q^d = Q^s$, so write that down and solve for P .

Line by line.

Line 1. $70 - 10P = 10P - 10$. The left side is demand, the right is supply, and setting them equal is the definition of equilibrium, not an assumption.

Line 2. $70 + 10 = 10P + 10P$. Add $10P$ to both sides to gather the price terms on the right; add 10 to both sides to gather the constants on the left.

Line 3. $80 = 20P$. Both sides collected.

Line 4. $P^* = 4$. Divide both sides by 20.

The star notation. P^* means the equilibrium price, as distinct from any old price P . The star marks the one value that makes the two sides agree.

No step above is beyond school algebra. That is worth saying plainly, because the reputation of economics suggests otherwise. The difficulty in this subject is almost never the manipulation — it is knowing which equation to write down in the first place.

Always check

Always check

Substitute back using **both** equations:

$$Q^d = 70 - 10(4) = 30, \quad Q^s = 10(4) - 10 = 30$$

Both give 30. That is your check.

Equilibrium: \$4 per cup, 30 thousand cups per week.

If they had disagreed, you would know to go looking.

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Working through it

Why substitute back at all. Solving gave us the price. The quantity has to be recovered separately, and doing it *twice* converts a calculation into a verified calculation.

Route one, through demand.

$$Q^d = 70 - 10(4) = 70 - 40 = 30$$

Route two, through supply.

$$Q^s = 10(4) - 10 = 40 - 10 = 30$$

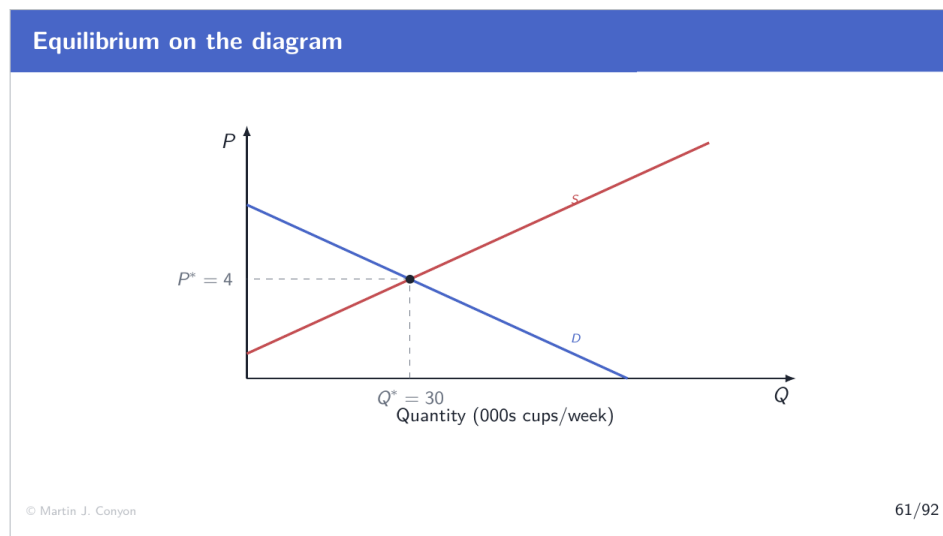
They agree, and they had to. A price at which the two *disagreed* would not be an equilibrium at all — that is precisely what we solved for.

So what is the check actually checking? Your arithmetic, not the economics. If line 3 above had been mis-collected, the two substitutions would return different numbers and you would know immediately. Without the check you would write down a wrong answer and never find out.

The result, stated with its units. \$4 per cup, 30 thousand cups per week. A price without units is not an answer.

Fifteen seconds. That is what the check costs, and it is the cheapest insurance available in this subject.

Equilibrium on the diagram



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Working through it

What is on the axes. Quantity horizontally, price vertically. The blue downward line is

demand, $Q^d = 70 - 10P$; the red upward line is supply, $Q^s = 10P - 10$.

Where the dot is and what it means. At the crossing point — the one place where the two curves agree about quantity. The dashed guides drop from it to \$4 on the price axis and across to 30 thousand on the quantity axis, matching the algebra exactly.

Why only one crossing. One curve falls and the other rises, so they meet once and only once. Two solutions would mean two equilibrium prices, which the straight-line model does not permit.

Where each curve begins and ends. Demand stops at its choke price of \$7; supply starts at its shut-down price of \$1. The equilibrium at \$4 sits inside that overlap, as it must — a crossing outside the range where both curves exist would not describe any actual trading.

One thing about how this figure was made. The dot was placed by the computer, which solved the two equations when the file compiled. Change the demand intercept in the source and the dot moves by itself. The picture cannot disagree with the arithmetic, because the picture *is* the arithmetic.

Why the market gets there

Why the market gets there

Shortage. $Q^d > Q^s$. Buyers want more than sellers will provide. Occurs when the price is *below* equilibrium.

Surplus. $Q^s > Q^d$. Sellers offer more than buyers will take. Occurs when the price is *above* equilibrium.

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Working through it

The question this frame opens. We have found a price where the two quantities match. But

nobody is in charge of the price. So why should a market sitting at some *other* price ever move to this one?

The two ideas needed to answer it.

Shortage. $Q^d > Q^s$. Buyers want more than sellers will provide. This happens when the price is *below* equilibrium, because a low price encourages buying and discourages producing.

Surplus. $Q^s > Q^d$. Sellers offer more than buyers will take. This happens when the price is *above* equilibrium, for the mirror-image reason.

Getting the direction right. Students reverse these. Anchor it on the behaviour rather than the word: a *low* price means eager buyers and reluctant sellers, which is a shortage. Work from the price to the behaviour and you will not have to remember which term goes with which.

What the next two frames do. Take one price below and one above, compute the gap, and watch what each gap does to the price. If both push toward \$4, we have an answer.

A shortage at \$3

A shortage at \$3

$$Q^d = 70 - 10(3) = 40, \quad Q^s = 10(3) - 10 = 20, \quad \text{shortage} = 20$$

Queues form. Sellers notice they can raise prices without losing customers — and they do.

As price rises, Q^d falls and Q^s rises.

The gap closes from both sides, and stops exactly at \$4.

Working through it

Where the two numbers come from. Substitute $P = 3$ into both equations.

$$Q^d = 70 - 10(3) = 40, \quad Q^s = 10(3) - 10 = 20$$

Buyers want 40 thousand; sellers offer 20 thousand. The shortage is the gap: $40 - 20 = 20$ thousand cups per week.

What that looks like on the ground. Queues. Shelves empty by lunchtime. Regulars turned away.

What sellers notice. That they could charge more and still sell everything they have. So they do — not by agreement, but individually, each responding to what is in front of them.

And here is the part worth pausing on: the gap closes from both ends. As the price rises, Q^d falls (movement along demand) and Q^s rises (movement along supply). Both curves are being climbed at once, from opposite directions.

Where it stops. At \$4, and nowhere else, because \$4 is the only price at which the gap is zero. Above \$3 the shortage shrinks; at \$4 it vanishes; and there is nothing left to push the price further.

Note what these movements are. Both are movements *along* curves, not shifts. Coffee's own price is what changed — exactly the test from Block 4.

A surplus at \$5

A surplus at \$5

$$Q^d = 70 - 10(5) = 20, \quad Q^s = 10(5) - 10 = 40, \quad \text{surplus} = 20$$

Unsold coffee. Sellers cut prices to shift it, Q^s falls, Q^d rises.

Again the market lands at \$4. It is self-correcting from either side.

Working through it

The mirror image, and the arithmetic is the same shape. Substitute $P = 5$:

$$Q^d = 70 - 10(5) = 20, \quad Q^s = 10(5) - 10 = 40$$

Sellers offer 40 thousand; buyers take 20 thousand. Surplus of 20 thousand cups.

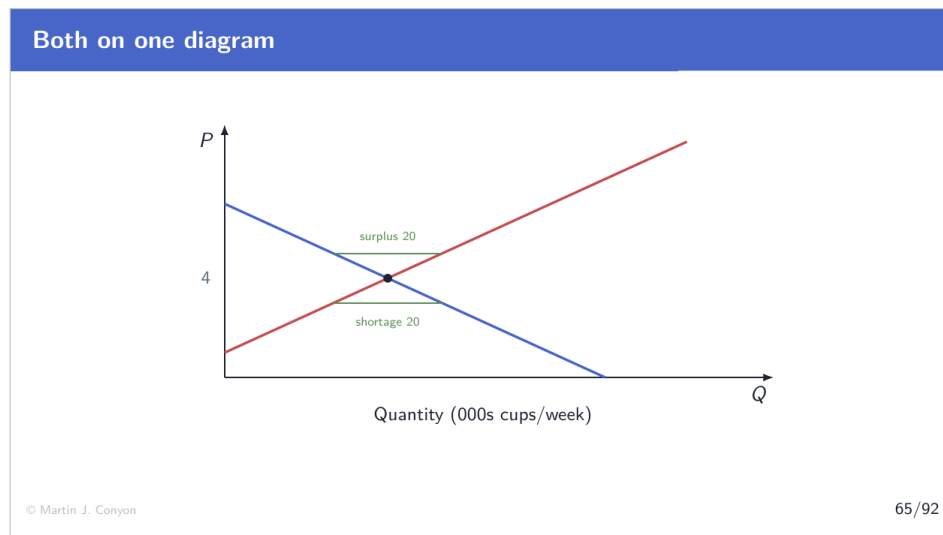
What that looks like. Unsold coffee. Stock going stale. Sellers cutting prices to shift it, again individually rather than by agreement.

And again the gap closes from both ends. As the price falls, Q^s falls and Q^d rises. The market lands at \$4 once more.

Why running the argument twice matters. One direction would only show that the market can *reach* equilibrium from below. Doing it from above shows the mechanism is **self-correcting from either side**, which is a much stronger claim.

And this answers the puzzle we opened the session with. Nobody organized the price of coffee. There is no committee. The price moves because individual buyers and sellers respond to shortages and surpluses in front of them, and it stops moving precisely when the market clears. The order is real, and it is nobody's plan.

Both on one diagram



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Working through it

What is on the axes. Quantity horizontally, price vertically, with the same two curves and the equilibrium dot at \$4.

The two green segments. The lower one, at \$3, is the shortage; the upper one, at \$5, is the surplus. Each spans horizontally from one curve to the other, because that horizontal distance *is* the gap between quantity supplied and quantity demanded at that price.

Where the endpoints came from. Not from a ruler. The lower segment runs from $Q^s = 20$ to $Q^d = 40$; the upper from $Q^d = 20$ to $Q^s = 40$. All four numbers were computed by substituting into the two equations, exactly as on the previous frames.

Why both segments happen to be 20 wide. Because \$3 and \$5 sit the same distance either side of \$4, and both curves are straight. That symmetry is a feature of these numbers, not a general rule.

The visual point. The gap narrows as you move toward the crossing point and vanishes there. Every price except \$4 generates pressure to move; \$4 generates none. That is what makes it an equilibrium rather than merely an intersection.

8. Part 8: Comparative Statics

What the model is actually for

What the model is actually for

Comparative statics. Comparing the equilibrium before an event with the equilibrium after it. It answers “what changed”, not “how long did it take”.

Everything so far has been machinery. **This is the part you use.**

Working through it

Why this block is the payoff. Blocks 1–7 built machinery: definitions, curves, a method for solving. None of it has yet answered a question anybody would ask outside a classroom. This block does.

Reading the definition. Comparative statics compares the equilibrium *before* an event with the equilibrium *after* it.

The word “statics” is doing work. We compare two resting points. We say nothing whatever about the journey between them.

So note carefully what the model does not tell you. It does not say how long adjustment takes, whether the price overshoots on the way, or what happens in the meantime. Those are genuinely harder questions.

Why that limitation is acceptable. Most practical questions are of the form “what will this do to the price?” rather than “what will the price be next Tuesday?”. Comparative statics answers the first kind, and it answers them with arithmetic rather than opinion.

The three-step method

The three-step method

Every question of this kind is answered the same way.

1. Decide **which curve shifts** — demand, supply, or both.
2. Decide **which direction**, and change the intercept accordingly.
3. **Solve** the new pair of equations. Read off P^* and Q^* .

Learn the method, not the answers.

Working through it

Three steps, and they never change.

Step 1 — which curve? Demand, supply, or both. Use the test from Block 4: if the event touches the good's own price, nothing shifts and you are moving along. Otherwise ask whose behaviour it changes. Something affecting buyers moves demand; something affecting sellers' costs moves supply.

Step 2 — which direction, and by how much? Change the intercept. A rightward shift for demand means a larger intercept; leftward, smaller. For supply, "more supplied at every price" also means a larger intercept, which moves the curve right and *down*.

Step 3 — solve. Set the new Q^d equal to the new Q^s and work through it exactly as in Block 7. Then substitute back — into both equations, as always.

Why learning the method beats learning the answers. There are only a few cases here, but there are unlimited possible events. An exam will hand you one you have never seen. The three steps will carry you through it; a memorised list of results will not.

Case 1: demand increases

Case 1: demand increases

A study reports that coffee is good for you. Demand rises by 20 at every price.

New demand	$Q^d = 90 - 10P$
Supply	$Q^s = 10P - 10$ (unchanged)
Set equal	$90 - 10P = 10P - 10$
	$100 = 20P$
	$P^* = 5$
Substitute	$Q^* = 10(5) - 10 = 40$

Price \$4 → \$5. Quantity 30 → 40. Both rise.

Working through it

Applying the three steps.

Step 1, which curve. A study about coffee's health effects changes what buyers want. It says nothing about production costs. So *demand* moves.

Step 2, which direction. Buyers want more at every price, so the curve shifts *right* and the intercept rises from 70 to 90. Supply is untouched at $10P - 10$.

Step 3, solve. Read down the table:

$$90 - 10P = 10P - 10 \implies 100 = 20P \implies P^* = 5$$

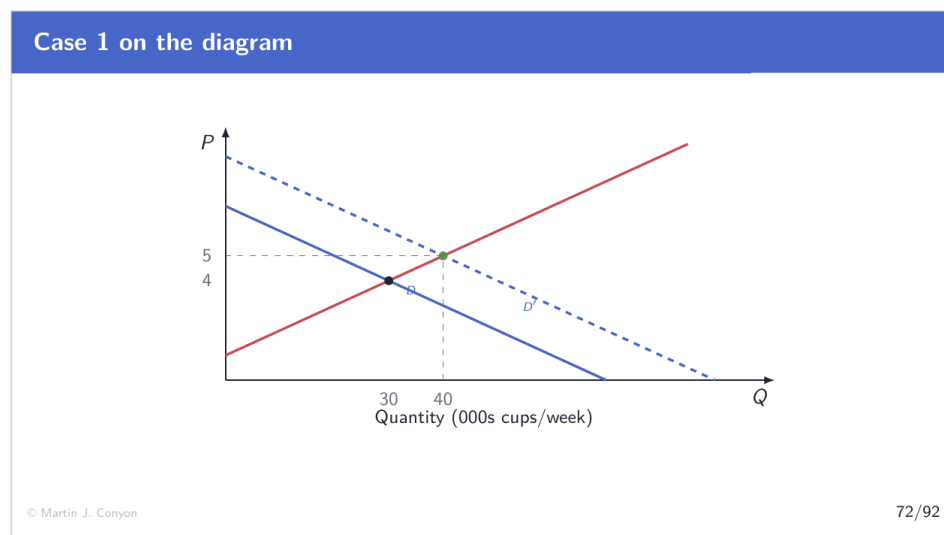
Then substitute back: $Q^* = 10(5) - 10 = 40$. And check with demand: $90 - 10(5) = 40$. Agreed.

Reading the result. Price rose from \$4 to \$5; quantity rose from 30 to 40. **Both up.**

Why both, intuitively. At every price buyers now want more than before, and that extra demand pressing on the existing supply bids the price up; the higher price then draws out more production. The market ends up trading more coffee at a higher price.

Hold on to this result. It is about to be compared with a case that looks identical on one measure and is not.

Case 1 on the diagram



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Working through it

What is on the axes. Quantity horizontally, price vertically, with two demand curves and one supply curve.

Which is which. D is the original demand, intercept 70. D' is the dashed new one, intercept 90, sitting to its right. The single red supply curve has not moved.

Reading the movement. The equilibrium travels from the lower dot to the upper-right one. Both coordinates increase: price \$4 to \$5, quantity 30 to 40, matching the algebra.

The observation worth extracting. The supply curve did not move, so the new equilibrium is simply a *different point on the same supply curve*. We slid *along* supply because demand shifted.

And that is a general pattern. When one curve shifts, the new equilibrium always lies on the curve that stayed still. Which is a useful check: if your answer does not sit on the unmoved curve, something has gone wrong.

Case 2: supply decreases

Case 2: supply decreases

A frost destroys part of the Brazilian crop. Supply falls by 20 at every price.

Demand	$Q^d = 70 - 10P$ (unchanged)
New supply	$Q^s = 10P - 30$
Set equal	$70 - 10P = 10P - 30$ $100 = 20P$ $P^* = 5$
Substitute	$Q^* = 10(5) - 30 = 20$

Price \$4 → \$5. Quantity 30 → 20. They move apart.

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Working through it

The same three steps, a different event.

Step 1. A frost destroys part of the crop. That affects producers' ability to supply, not buyers' willingness to buy. So *supply* moves.

Step 2. Less is available at every price, so the curve shifts *left* and the intercept falls from -10 to -30 . Demand is untouched.

Step 3.

$$70 - 10P = 10P - 30 \implies 100 = 20P \implies P^* = 5$$

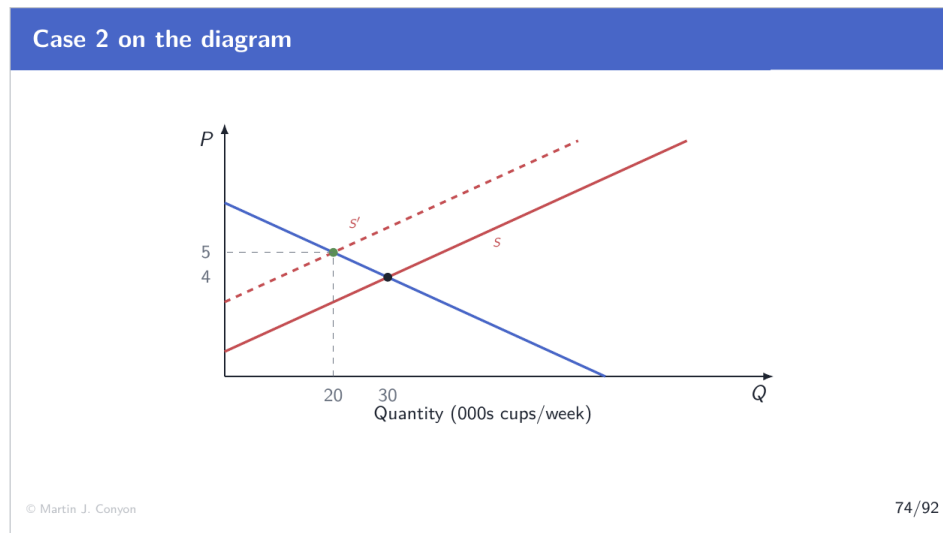
Substitute back: $Q^* = 10(5) - 30 = 20$. Check with demand: $70 - 50 = 20$. Agreed.

Reading the result. Price rose from \$4 to \$5 — and quantity *fell*, from 30 to 20. They moved in **opposite** directions.

Why, intuitively. Coffee has become harder to produce, so less of it is made; the scarcity bids the price up. Less coffee, more expensive.

Now compare that with Case 1, because something has happened that is worth two more frames.

Case 2 on the diagram



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Working through it

Which is which. S is the original supply curve; S' is the dashed new one, up and to the

left. The blue demand curve has not moved.

Reading the movement. The equilibrium travels up and to the left: price \$4 to \$5, quantity 30 to 20.

Why leftward means upward here. A leftward shift of supply is also an *upward* one — sellers now require a higher price for any given quantity. The two descriptions are the same movement seen along different axes, which is worth being comfortable with.

The pattern from Case 1, confirmed. This time demand stayed still, so the new equilibrium lies on the *demand* curve. We slid along demand because supply shifted.

So the general rule holds both ways. The equilibrium always ends up on the curve that did not move. Two curves shifting at once — Case 3 — is the exception, and it behaves differently for exactly that reason.

Compare the two cases

Compare the two cases

Both raised the price to exactly \$5.

A student who *only looked at the price* could not tell them apart.

The quantity distinguishes them: it rose in Case 1, fell in Case 2.

Price alone never identifies which curve moved.

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Working through it

Put the two results side by side.

Case 1: price \$4 → \$5, quantity 30 → 40.

Case 2: price \$4 → \$5, quantity 30 → 20.

The prices are identical. Not similar — identical. Two completely different events, a health study and a frost, produced exactly the same equilibrium price.

So consider what an observer sees. You read that coffee now costs \$5, having cost \$4. From that fact alone you cannot tell whether buyers wanted more coffee or producers could supply less. The two stories have opposite implications for everyone involved, and the price does not distinguish them.

What does distinguish them. The quantity. It rose in Case 1 and fell in Case 2. Observe both price and quantity and the ambiguity disappears.

The general statement. **Price alone never identifies which curve moved.** That is not a fact about this example; it is a fact about the model, and it is the beginning of a genuinely deep problem in economics — inferring causes from observed outcomes when several causes produce the same outcome.

Case 3: both curves shift

Case 3: both curves shift

Demand rises by 20 and supply falls by 20.

$$90 - 10P = 10P - 30 \Rightarrow 120 = 20P \Rightarrow P^* = 6, \quad Q^* = 30$$

Price rises to \$6. Quantity is 30 — **exactly where it started.**

Both shifts push price the same way, so price is unambiguous. They push quantity in *opposite* directions, so quantity depends on which shift is bigger.

Working through it

Both events at once. Demand up 20 (intercept $70 \rightarrow 90$) and supply down 20 (intercept $-10 \rightarrow -30$). No new technique: same three steps.

Solving.

$$90 - 10P = 10P - 30 \implies 120 = 20P \implies P^* = 6$$

and $Q^* = 10(6) - 30 = 30$. Check with demand: $90 - 60 = 30$. Agreed.

The striking result. The price rose to \$6, higher than in either case alone. But the quantity is 30 — **exactly where it started**.

Why the price is unambiguous. Both shifts push it the same way. Higher demand raises the price; lower supply raises the price. Two forces, same direction, so the answer cannot be in doubt — and the effect is larger than either alone.

Why the quantity is not. They push it in *opposite* directions. Higher demand pulls quantity up; lower supply pushes it down. Which wins depends entirely on their relative sizes.

And here they cancel exactly. That is a property of the numbers chosen, not a law. Make the demand shift 30 instead of 20 and quantity would rise; make the supply shift larger and it would fall.

The general rule. When both curves move, one of price and quantity is determinate and the other is ambiguous, and no diagram can resolve the ambiguous one. Only the numbers can.

9. Part 9: From Table to Equation to Graph

We have been doing this backwards

We have been doing this backwards

All lecture, I gave you an equation and you produced the table and the graph.

Nobody hands you an equation.

Data arrives as a **table**: prices, and the quantities observed at those prices.

The equation is something *you* have to find.

Working through it

What this block reverses. Every earlier block began with an equation. I supplied it, and you produced the table and the graph.

Why that is the wrong way round. Equations do not exist in nature. Nobody observes $Q^d = 70 - 10P$. What is actually observed is transactions: at this price, this many were bought; at that price, that many.

So the real sequence runs the other way. Data arrives as a table. The equation is the thing you have to *find*.

And once found, it does something the table cannot. It answers questions about prices nobody ever observed — which is the whole reason for building it.

The four steps, previewed. Find the slope from any two rows. Find the intercept by substituting one row. Check against a row you did not use. Then plot, and solve. Each gets its own frame.

The data

The data

A market for bicycle helmets. Observed at five prices:

Price P (\$)	1	2	3	4	5
Quantity demanded Q^d (000s/week)	90	80	70	60	50
Quantity supplied Q^s (000s/week)	0	20	40	60	80

That is all you get. **Four steps get you from here to a full model.**

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Working through it

What you have, and what you do not. Ten numbers: five prices, with the quantity demanded and quantity supplied observed at each.

You do not have an equation, a curve, a slope, an intercept, a choke price, a shut-down price, or an equilibrium. All of those are about to be extracted from these ten numbers.

Read the two rows in opposite directions. Quantity demanded falls as the price rises — 90, 80, 70, 60, 50. Quantity supplied rises — 0, 20, 40, 60, 80. The two laws are visible in the data before any theory is applied to it.

And one row is already interesting. At $P = 4$, both quantities are 60. Hold that thought; it will matter at the end, and so will the fact that it is a coincidence of which prices happened to be recorded.

Why bicycle helmets rather than coffee. Because the coffee numbers are familiar by now, and the point of this block is to run the method on something you have not already memorised.

Step 1 — find the slope: rise over run

Step 1 — find the slope: rise over run

Pick any two rows of the demand data. Take $P = 1$ and $P = 2$:

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\text{change in } Q^d}{\text{change in } P} = \frac{80 - 90}{2 - 1} = \frac{-10}{1} = -10$$

Each extra dollar reduces quantity demanded by 10 thousand helmets.

The minus sign is the law of demand, arriving from the data rather than from theory.

Working through it

The formula, in the form you already know it. Slope is rise over run: the change in the vertical quantity divided by the change in the horizontal one. Here, the change in Q^d divided by the change in P .

Building it from the first two rows. Quantity goes from 90 to 80, so the rise is $80 - 90 = -10$. Price goes from 1 to 2, so the run is $2 - 1 = 1$.

$$\text{slope} = \frac{-10}{1} = -10$$

Reading what that means. Each extra dollar of price costs 10 thousand helmets. That is the same statement the coefficient made in every earlier block, arrived at from data instead of being handed over.

And the minus sign is worth dwelling on. It did not come from theory. Nobody told these buyers about the law of demand. It fell out of ten observed numbers — which is where an empirical law ought to come from.

One caution about direction. Subtract in a consistent order. Here we went from the $P = 1$ row to the $P = 2$ row in both numerator and denominator. Mixing the order flips the sign and produces nonsense.

Any two rows give the same answer

Any two rows give the same answer

Rows used	Rise over run	Slope
$P = 1$ and $P = 2$	$(80 - 90)/(2 - 1)$	-10
$P = 2$ and $P = 4$	$(60 - 80)/(4 - 2)$	-10
$P = 1$ and $P = 5$	$(50 - 90)/(5 - 1)$	-10

The same slope every time. That is what makes the relationship linear.

If different pairs of rows gave different slopes, the curve would be bent — and a straight-line equation would be the wrong model.

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Working through it

Why check other pairs at all. Because the answer depended on which two rows were picked, and if it varies, the whole approach is wrong.

Three pairs, worked.

$$\frac{80 - 90}{2 - 1} = -10, \quad \frac{60 - 80}{4 - 2} = \frac{-20}{2} = -10, \quad \frac{50 - 90}{5 - 1} = \frac{-40}{4} = -10$$

Adjacent rows, rows two apart, and the two extreme rows. All -10 .

What that constancy proves. The relationship is **linear**. A constant slope everywhere is precisely what a straight line means, so fitting a straight line is legitimate.

What non-constancy would have meant. Suppose the pairs had given -10 , -14 and -19 . The underlying curve would be *bent*, and fitting a straight line would produce an equation that fits the two rows used and misses everywhere else.

So this is not busy work. It is the step that tells you whether the model you are about to build is the right *kind* of model. Skipping it means never finding out.

Step 2 — find the intercept

Step 2 — find the intercept

We know the shape: $Q^d = a - 10P$. One unknown left: a .

Substitute any row you like. Take $P = 1$, $Q^d = 90$:

$$90 = a - 10(1)$$

$$90 = a - 10$$

$$a = 100$$

So $Q^d = 100 - 10P$.

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Working through it

Where we are. We know the shape — $Q^d = a - 10P$ — because step one gave us the slope. One unknown remains, so one equation will pin it down, and any row of the table supplies one.

The substitution, line by line. Take the $P = 1$ row, where $Q^d = 90$.

$90 = a - 10(1)$ — substitute both values into the shape.

$90 = a - 10$ — evaluate the product.

$a = 100$ — add 10 to both sides.

So $Q^d = 100 - 10P$.

Any row would have done. Try $P = 4$, $Q^d = 60$: $60 = a - 40$, so $a = 100$. Same answer, as it must be if the relationship really is linear — which step one established.

And now notice what has just happened. Nobody ever sold a helmet at a price of zero. That intercept was never observed by anyone. We **inferred** it from five observations, and that inference is the first thing the model gives us that the data could not.

Step 3 — check against a row you did not use

Step 3 — check against a row you did not use

We built the equation from the $P = 1$ row. So test it on a different one.

At $P = 4$: $Q^d = 100 - 10(4) = 60$. Table says **60**. ✓

At $P = 5$: $Q^d = 100 - 10(5) = 50$. Table says **50**. ✓

Never accept an equation you fitted without testing it on data you did not use to fit it.

Working through it

Why testing against the $P = 1$ row would prove nothing. We used that row to find the intercept, so the equation is guaranteed to reproduce it. Fitting a line through a point and then observing that the line passes through the point is not evidence.

So test rows we did not use.

At $P = 4$: $Q^d = 100 - 10(4) = 60$. The table says 60. ✓

At $P = 5$: $Q^d = 100 - 10(5) = 50$. The table says 50. ✓

Now the equation has earned some trust. It reproduces data it was not built from, which is a genuinely different claim from reproducing data it was.

The principle, stated generally. Never accept an equation you fitted without testing it against a row you did not use to fit it. Here that confirms the arithmetic and that the listed points really are collinear. It is not evidence that the same straight line holds outside the table.

Where this goes. That principle scales all the way up. In serious empirical work it acquires formal names and procedures — holding back part of a sample, testing predictions out of sample — but the logic is exactly what you have just done by hand with two rows of a table.

Now do the same for supply

Now do the same for supply

Slope. From $P = 1$ and $P = 2$: $\frac{20 - 0}{2 - 1} = +20$ — positive, as the law of supply requires.

Intercept. $Q^s = c + 20P$. At $P = 1$, $Q^s = 0$:

$$0 = c + 20(1) \implies c = -20$$

So $Q^s = 20P - 20$.

Check at $P = 5$: $20(5) - 20 = 80$. Table says **80**. ✓

Working through it

Three steps again, compressed, because you have seen them.

Slope. From the $P = 1$ and $P = 2$ rows: quantity goes from 0 to 20, price from 1 to 2.

$$\frac{20 - 0}{2 - 1} = +20$$

Positive, as the law of supply requires — and again arrived at from data rather than assumed.

Intercept. The shape is $Q^s = c + 20P$. Substitute the $P = 1$ row, where $Q^s = 0$:

$$0 = c + 20(1) \implies c = -20$$

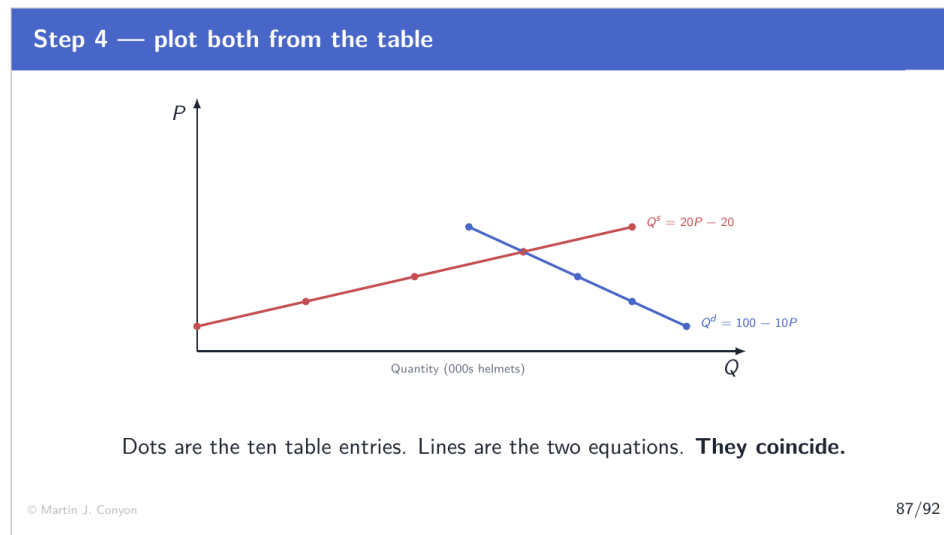
So $Q^s = 20P - 20$.

Check, on a row not used. At $P = 5$: $20(5) - 20 = 80$. The table says 80. ✓

And now look at that negative intercept, because you have met one before. Set $Q^s = 0$: $20P - 20 = 0$, so $P = 1$. The shut-down price is \$1.

Which the data confirms independently. The table shows zero supplied at exactly \$1. The algebra and the observations agree — and note that they are two separate sources of that fact, not one.

Step 4 — plot both from the table



Slide 87

Working through it

What is on the axes. Quantity horizontally, price vertically, as throughout.

What the marks are. Ten dots: five blue for the demand observations, five red for supply. Two lines: the fitted equations $Q^d = 100 - 10P$ and $Q^s = 20P - 20$.

Why they coincide. Because the lines were built from those dots. If a dot sat off its line, either the arithmetic is wrong or the relationship is not linear.

Which way each slopes. Demand falls, supply rises — the two laws, drawn.

Now look at the ends of the lines. They extend *beyond* the dots, into regions where no data exists.

That extension is the whole point of the exercise. The equation makes claims about prices nobody observed: what would be demanded at \$8, at what price supply dries up entirely. A table can say nothing about a price not in it. A model can, and the next frame is about exactly what it says.

Step 5 — equilibrium, two ways

Step 5 — equilibrium, two ways

From the table. Look down the two rows for a price where the numbers match. At $P = 4$: $Q^d = 60$ and $Q^s = 60$.

From the equations.

$$100 - 10P = 20P - 20$$

$$120 = 30P$$

$$P^* = 4, \quad Q^* = 100 - 10(4) = 60$$

Both routes give \$4 and 60 thousand. **Two methods, one answer.**

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Working through it

Route one, from the table. Read down the two rows looking for a price where the quantities

match. At $P = 4$: $Q^d = 60$ and $Q^s = 60$. Equal, so that is the equilibrium.

Route two, from the equations.

$100 - 10P = 20P - 20$ — set the two equal, which is the definition of equilibrium.

$120 = 30P$ — collect the constants left, the price terms right.

$P^* = 4$ — divide by 30.

Then substitute back: $Q^* = 100 - 10(4) = 60$. And check with supply: $20(4) - 20 = 60$. Agreed.

Two routes, one answer. They share no step whatever. One reads a table; the other manipulates algebra. Both give \$4 and 60 thousand.

Why the agreement is worth something. If the equations had been fitted wrongly, they would not reproduce the table's own equilibrium. The check tests the whole chain — slope, intercepts, and solving — in one step.

Why the equations beat the table

Why the equations beat the table

The table gave us the equilibrium here only because \$4 *happened* to be one of the five rows.

It usually will not be.

The equations work at every price, observed or not. They give you the choke price of \$10 — which appears nowhere in the data, and follows only from extending the fitted line past the prices we were given.

That is what you gain by turning a table into a model.

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Working through it

The uncomfortable observation. The table gave us the equilibrium only because \$4 *happened* to be one of the five prices somebody recorded.

And usually it will not be. Change the observed prices to \$1, \$3 and \$5 and the equilibrium at \$4 appears in no row at all. The table would then be silent, and the equations would answer immediately. The Active Learning exercise that follows is built on exactly this point.

What else the equations know.

The choke price. Set $Q^d = 0$: $100 - 10P = 0$, so $P = 10$.

The shut-down price. Set $Q^s = 0$: $20P - 20 = 0$, so $P = 1$.

One of these is in the table and one is not. The \$1 shut-down price appears directly — it is the row where $Q^s = 0$. The \$10 choke price does not: the highest price recorded was \$5, and nobody observed a market at \$10. It follows only by **extending the fitted line** beyond the listed prices, which is valid only if the straight line really does continue.

That is what turning a table into a model buys you. Not a tidier description of what was seen — claims about what was not. And it is what the whole session has been building toward: the four steps in Blocks 2 and 5 taught you to run the machine forwards, and this block shows you how to build the machine in the first place.